

General selection criteria for Wi-Fi and RF modules

Module selection criteria are a case-by-case process. The one-size-fits-all approach does not work here. Module selection is based on the following factors:

Deployment scenario. What are the obstacles present for the solution? Modules installed inside houses would have maximum number of obstacles, while those installed on the streets would have the least. Geography would also play a part, as Asian countries would see more concrete based obstacles that are difficult to penetrate, while North American and European countries would have wood based obstacles that are relatively easier to penetrate.

Range. What is the range needed for the modules? If the modules are placed far away from each other, the technology to be selected needs to have a high near-line-of-sight range. Modules placed closer to each other can afford to have lower range.

Network. Radio modules are made to work in either a mesh network or a star network, most of the times depending upon how the modules are to be placed. When multiple modules are placed close to each other and the data concentrator unit/remote terminal unit is far away, a mesh network is preferred. On the other hand, if data collection units are closer and are more in number, then a star network can be employed.

Existing infrastructure or new infrastructure. Existing infrastructure like utility meters and lighting fixtures need to be considered. Completely changing existing devices is expensive and, most of the times, not feasible. As a result, radio modules need to be able to work with these devices with little or no change to these. This can happen by ensuring that radio modules can communicate with existing devices using protocols like RS232 and RS485. On the other hand, if new devices are to be installed, then the radio modules need to be pre-fitted into these and, as a result, must have production capabilities.

Anurag Ambedkar, vice president - international business, Maven Systems Pvt Ltd

Wi-Fi. Multitech MTDOT-868-X1-SMA is another LoRa Wi-Fi. USR is a Chinese company with plenty of solutions and viral among its solution is USR-WIFI232-T. ESP8266 by Expressive has a large variety of modules from ESP01 to ESP14 that dominate the market. Some of the latest modules are ESP8066 12E, 14 and Wroom and Esp32 Wi-Fi and Bluetooth, XBee Wi-Fi, node MCU and Adafruit cc3000."

T. Anand, managing director, Knewron Technologies, says "For RF-driven modules, nRF24L01+ and RFM-96 are well-known modules. Both have closely-matching criteria."

RajGuru Electronics offers XBee 2mW wire antenna - series 2, XBee 1mW wire antenna - series 1, XBee Pro 63mW wire antenna - series 2B (ZigBee Mesh), XBee Explorer USB, XBee Explorer - Serial. Their Bluetooth modules (HC-05, HC05-TTL, ELM327, HM Bluetooth module and Bluetooth 4.0) offer an option of slave-and-master mode to the user and are available in a variety of specifications at reasonable prices.

Interface. RF modules typically communicate with an embedded system such as an MCU or micro-processor. Communication protocols include UART used in Digi International's XBee modules, Serial_Peripheral_Interface_Bus used in Anaren's AIR modules and USB used in Roving Networks' modules. Although the modules may use a standardised protocol for wireless communication, commands sent over the MCU interface are typically not standardised, as each vendor has its own proprietary communications format.

Speed of the MCU interface depends on the speed of the underlying RF protocol used—higher speed RF protocols such as Wi-Fi require a high-speed serial interface such as USB, whereas protocols with a slower data rate such as Bluetooth Low Energy may use a UART interface.

IoT-driven modules. Embedded electronic devices are now adding the capability to be accessible from the Internet or be able to push data to the Cloud. This new trend in connecting smart electronic devices to

the Internet is called the IoT. Many research studies have envisioned that 20 to 50 billion smart devices or things will be connected to the Internet by 2020. Attractiveness of the IoT market is driving the creation of many new businesses.

Main hardware components of an IoT device are:

Microcontroller unit. The processing unit together with the memory device

Connectivity module. Wireless or wired connectivity module enabling the device to communicate with other devices

Sensor module. Temperature, proximity, movement, humidity, current, voltage and so on

As in any design activity, there are several approaches to designing IoT hardware and devices like chip design solution, SoC design solution and embedded module solution. To avoid the risk of failure and project delays, many companies go with the module path. That does not mean that they are engaged in an easy path.

Wireless module selection is another challenge. Ambedkar says, "IoT-driven Wi-Fi modules connect a sensor or end node device directly to the Internet. These require a router in order to upload collected data to the Internet. Issues faced by such modules usually include limited penetration power, inefficient mesh network and lack of a standard protocol like ZigBee. On the other hand, these modules are relatively cheaper and can be accessed using mobile phones and tablet PCs, thus giving a better option for user touch points.

"IoT-driven RF modules connect to a gateway device and transmit data to it. The gateway then uses an appropriate connection backbone like GPRS, Wi-Fi, WiMAX or Ethernet in order to upload data to the Internet. Issues faced by these modules include an additional intermediary to upload data like a gateway and higher cost than Wi-Fi modules.